

# In Situ Characterization of Lifetime and Morphology in Operating Bulk Heterojunction Organic Photovoltaic Devices by Impedance Spectroscopy

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Time-dependent charge transport in operating poly(3-hexylthiophene):[6,6]-phenyl-C<sub>61</sub>-butyric acid methyl ester (P3HT:PCBM) bulk heterojunction organic photovoltaic (OPV) devices has been characterized with impedance spectroscopy. Devices with varied composition and morphology were measured over a range of illumination intensities ranging from dark conditions to 1 sun and applied bias voltages ranging from 0.0 V to 0.75 V. Using an equivalent circuit model, materials properties such as dielectric constant and conductivity were determined and found to be in agreement with values measured by other methods. Average carrier lifetimes were also extracted from the model and found to correlate with measured power conversion efficiencies. At the short circuit condition and ~1 sun illumination, the average electron lifetime was found to vary from 7.8 to 22  $\mu$ s for devices with power conversion efficiencies ranging from 2.0 to 2.5%. These results suggest that impedance spectroscopy is an effective tool for predicting how processing parameters can impact device performance in organic bulk heterojunction photovoltaic devices.

## 1. Introduction

Bulk heterojunction (BHJ) organic photovoltaic devices offer the potential for flexible, light-weight, low-cost solar cells manufactured by scalable methods such as roll-to-roll processing and

spray-coating.<sup>[1–5]</sup> Understanding the relationship between processing parameters and morphology has been essential in facilitating efficiency improvements in these devices. In donor-acceptor systems such as poly[2-methoxy-5-(3',7'-dimethyloctyloxy)-1,4-phenylene vinylene]:PCBM (MDMO-PPV:PCBM),<sup>[6]</sup> P3HT:PCBM<sup>[7,8]</sup> and polythienothiophene-co-benzodithiophene:PCBM (PTB:PCBM),<sup>[9,10]</sup> for example, a strong correlation has been observed between device performance and processing factors such as solvent choice and annealing conditions. In the ubiquitous P3HT:PCBM system, drying time during the solvent-annealing step markedly impacts device performance, with power conversion efficiencies varying from 1.36% in the case of a 20 s anneal time to 4.37% when the film is annealed for 20 min.<sup>[7]</sup> This effect is attributed to mor-

phology enhancements induced by the slower annealing rate.

To elucidate the impact of processing parameters on the nanoscale morphology of BHJ OPV active layers, numerous tools have been employed including transmission electron microscopy/tomography,<sup>[11,12]</sup> scanning electron microscopy,<sup>[13]</sup> and scanning

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probe techniques such as atomic force microscopy (AFM),<sup>[14]</sup> Kelvin probe microscopy,<sup>[15]</sup> photoconductive AFM,<sup>[16]</sup> and atomic force photovoltaic microscopy.<sup>[17]</sup> Spectroscopic ellipsometry<sup>[18]</sup> and X-ray diffraction<sup>[19]</sup> have also yielded insights into morphological evolution, including the role and extent of vertical phase segregation. Other techniques have provided indirect information about active layer morphology by enabling the determination of charge carrier lifetimes. These approaches have included photo-induced charge carrier extraction by linearly increasing voltage (photo-CELIV),<sup>[20]</sup> transient absorption spectroscopy (TAS),<sup>[21,22]</sup> transient photovoltage measurements,<sup>[23]</sup> and time-of-flight<sup>[24]</sup> techniques. While these methods have proven to be quite informative, the relationships between nanoscale morphology, device performance, and the underlying electrical characteristics of functioning devices are not yet fully understood.

Impedance spectroscopy is a powerful, non-destructive characterization tool that is commonly used for the analysis of a wide variety of electrical devices. For dye-sensitized solar cells (DSSCs), it has become an established method for determining charge carrier lifetimes under a range of operating biases and illumination intensities.<sup>[25–30]</sup> DSSCs have been successfully modeled by equivalent circuit elements, which have helped to elucidate the roles of internal interfaces as well as device components such as the liquid electrolyte and the transparent conductor. The two key parameters extracted from these equivalent circuits are the recombination resistance and the chemical capacitance, which together allow determination of average charge carrier lifetime.

Recently, impedance analysis has begun to be applied to BHJ organic photovoltaic devices.<sup>[31,32]</sup> These reports demonstrate that the impedance analysis framework established for DSSCs can, to some extent, be extended to bulk heterojunction devices in order to calculate average charge carrier lifetime, electronic densities-of-states, and charge carrier concentrations. However, a detailed analysis of the impedance response of active devices, and its dependence on device processing history, morphology, and operating conditions is still needed. In this work we acquire and analyze the impedance behavior of P3HT:PCBM bulk heterojunction devices and its dependence on illumination intensity and bias conditions, active layer composition, and annealing history. We also report a simplified equivalent circuit model that successfully describes bulk heterojunction devices over a range of illumination conditions and applied biases. We use this model to extract relevant device performance characteristics and find, in agreement with other studies employing complementary techniques,<sup>[33–36]</sup> that bimolecular recombination losses play a significant role in the performance of these devices. To this end, we demonstrate a correlation between device efficiency and lifetime, and describe how values extracted from the present equivalent circuit model can be used to optimize device performance with new materials systems.

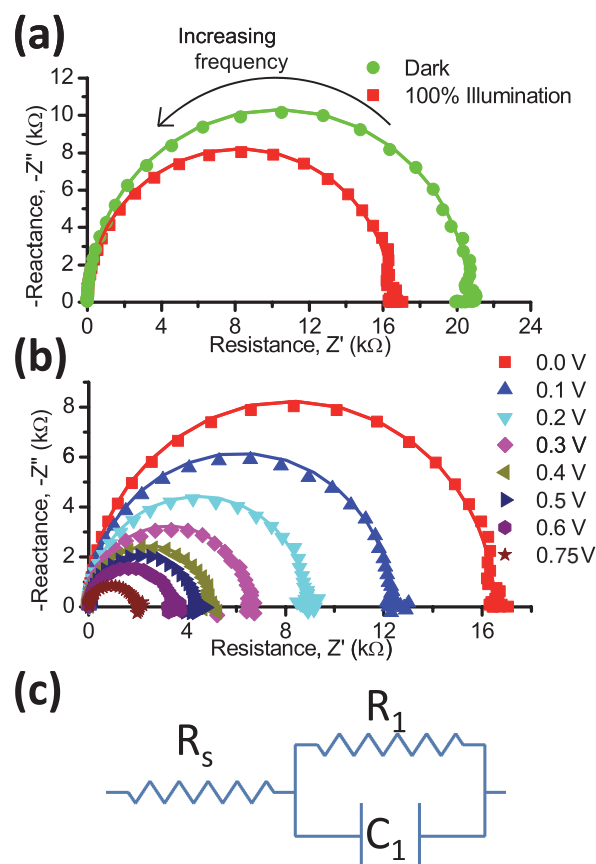
## 2. Results and Discussion

### 2.1. P3HT-Only Devices

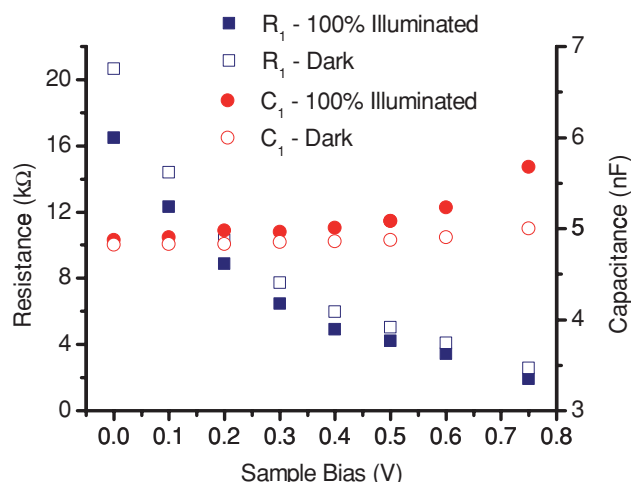
In order to provide a basis for comparison to the complete P3HT:PCBM BHJ system, devices consisting of a P3HT-only

active layer were first examined. The structure of the devices was ITO/PEDOT:PSS/P3HT/Al, and as expected, they exhibited only a small photovoltaic response, with a power conversion efficiency (PCE) of about 0.0040%. Cole-Cole plots of a P3HT-only device under illuminated and dark conditions are shown in Figure 1(a), while the response to a large range of DC applied biases, under illumination, is shown in Figure 1(b). These plots are complex plane representations of the imaginary part of the impedance response, reactance or  $Z''$ , and the real part of the impedance, resistance or  $Z'$ . The devices exhibit a single semicircle in the complex plane whose diameter is highly dependent on both illumination and bias. All of the semicircles intersect the  $Z'$  axis at a value of  $\sim 15 \Omega$  at high frequencies and again at low frequencies at a value that depends on the testing conditions.

These data were modeled by the simple equivalent circuit shown in Figure 1(c), consisting of a resistor  $R_s$  in series with a resistor  $R_1$  and capacitor  $C_1$  in parallel. It is well known that a parallel  $R$ - $C$  combination gives rise to a semicircle in the impedance plane,<sup>[37]</sup> and as depicted in Figure 1, the model provides an excellent fit for the data both under illuminated and dark conditions as well as over a range of applied biases. The  $R_s$  circuit element represents resistive losses in the ITO and



**Figure 1.** Impedance Response of P3HT-only device. (a) Device under illuminated and dark conditions at 0.0 V. The points represent the measured data, and the lines are the fit. (b) Device under varied sample bias at 1 sun illumination. (c) Equivalent circuit used to fit the data in (a) and (b).  $R_s$  represents the resistive losses in the ITO and PEDOT:PSS, and  $R_1$  and  $C_1$  represent the response of P3HT.



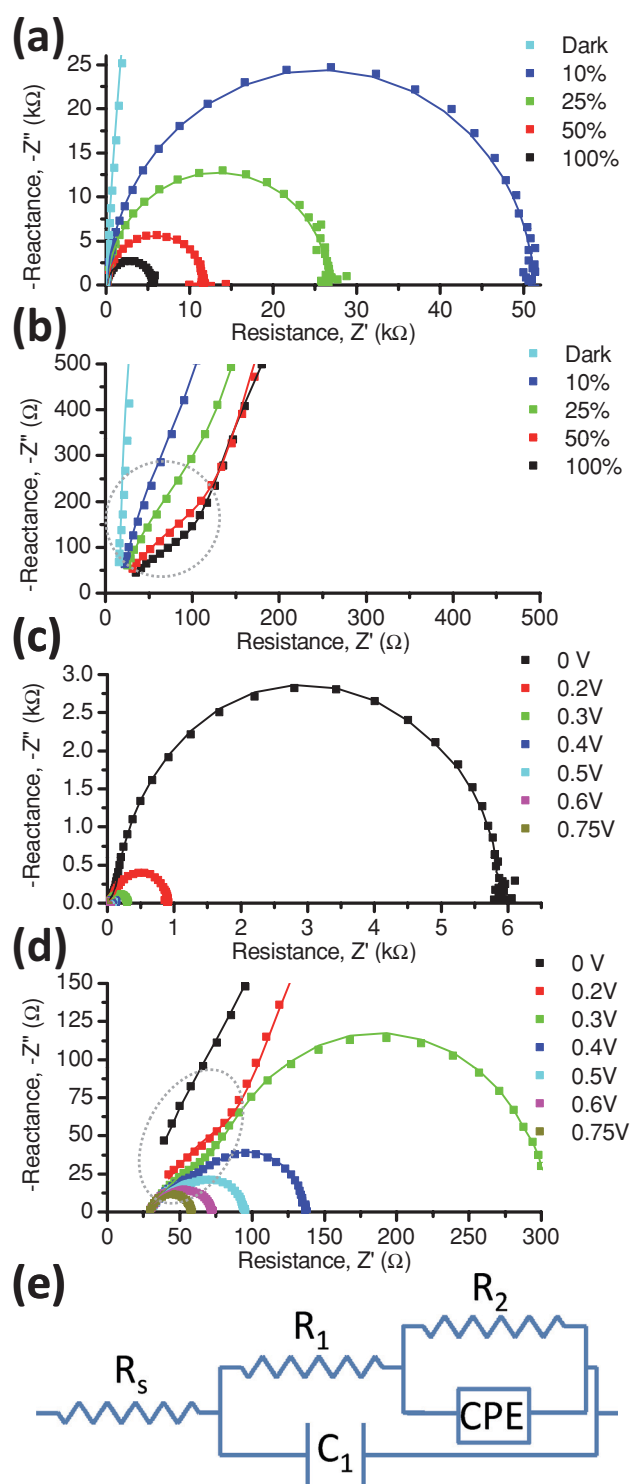
**Figure 2.** Response of  $R_1$  and  $C_1$  to varied sample bias.  $R_1$  decreases with increasing bias for both dark and illuminated conditions while  $C_1$  is relatively insensitive to changes in bias.

PEDOT:PSS and corresponds to the intersection of the semicircle with the  $Z'$  axis at high frequencies. It is nearly constant under the illumination and sample bias conditions examined here, with a value of  $\sim 15 \Omega$ , as expected. The resistance  $R_1$  corresponds to the diameter of the semicircle and decreases in response to illumination or increasing sample bias as shown in Figure 2. It is attributed to the bulk resistance of the P3HT film, which is reduced by the introduction of charge carriers either through illumination (i.e., photoconductivity) or electrical injection at the electrodes.

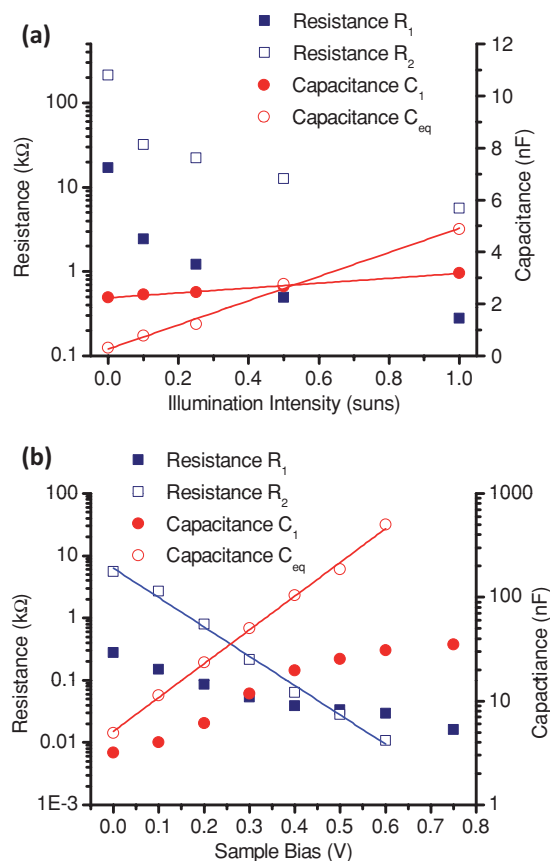
The intrinsic conductivity of P3HT, as determined from the value of  $R_1$  at 0.0 V and the layer dimensions, was found to be  $2.7 \times 10^{-9} \text{ S/cm}$  in the dark and  $3.4 \times 10^{-9} \text{ S/cm}$  under 1 sun illumination. These values increase to  $2.2 \times 10^{-8} \text{ S/cm}$  and  $2.9 \times 10^{-8} \text{ S/cm}$  at 0.75 V ( $8.1 \text{ V}/\mu\text{m}$ ), a change of approximately one order of magnitude, in response to the electrical injection of charge carriers. The conductivity of P3HT is known to be influenced by its orientation on the substrate, degree of regioregularity, as well as exposure to air,<sup>[38,39]</sup> and a Schottky barrier at either electrode could additionally influence  $R_1$ . Nonetheless, the values measured here are similar to previously reported work.<sup>[40]</sup> The capacitance  $C_1$  is simply the geometrical capacitance of the device. As shown in Figure 2, it has a value of about 4.9 nF that is essentially independent of both illumination and bias except at large voltages. Assuming the equation for a simple parallel plate capacitor, this translates into a calculated dielectric constant ( $\epsilon$ ) of 3.1 using the measured film thickness of 93 nm and device area of  $0.165 \text{ cm}^2$ . This result is consistent with previously reported values.<sup>[41]</sup> The slight increase in capacitance at larger voltages is consistent with previous work on organic blend photodetectors.<sup>[42]</sup> It is likely associated with an increase in the effective permittivity due to the injected charge carriers under illumination and higher applied biases, however it is not yet fully understood.

## 2.2. Bulk Heterojunction P3HT:PCBM Devices

The impedance results for a BHJ device having the structure ITO/PEDOT:PSS/P3HT:PCBM(1:1)/Al are shown in Figure 3 and 4.



**Figure 3.** Impedance Response of P3HT:PCBM (1:1) device. (a) Device at 0.0 V sample bias and varied illumination intensity. 100% illumination  $\approx 1$  sun. (b) High frequency portion of 3(a). Note that x-axis has been scaled to increase curve separation. (c) Device under 1 sun illumination and varied sample bias. (d) Low resistance/reactance region of 3(c) expanded to show behavior at higher sample biases. (e) Equivalent circuit for P3HT:PCBM device. Note that in (b) and (d) the portions of the spectra exhibiting the high frequency partial arcs are indicated by the dotted ellipsoids.



**Figure 4.** Response of  $R_1$ ,  $R_2$ ,  $C_1$ , and  $C_{eq}$  to illumination and sample bias variation for the control device. (a) Response of circuit elements to illumination variation at 0.0 V. (b) Response of circuit elements to bias variation for the device illuminated at 1 sun.  $R_1$  and  $C_1$  dominate the impedance response at biases  $> 0.6$  V. Lines are linear (a) and exponential (b) fits to the data.

The spectra exhibit a large semicircle that dominates the response over most frequencies as shown in Figures 3a and 3c. At high frequencies a second partial arc is clearly visible (Figures 3b and 3d), which becomes more easily resolved at high illumination intensities and applied biases in agreement with previous work by Garcia-Belmonte and coworkers.<sup>[32]</sup> In Figure 3a the diameter of the large semicircle is significantly larger than that for the P3HT-only device in the dark. In addition, the response of this semicircle to both illumination intensity and to applied bias is more sensitive than that of the P3HT-only device, a fact we attribute to internal charge transfer events at the donor/acceptor interfaces. At low frequencies, the AC impedance response approaches the DC resistance and a correlation can be made between the Cole-Cole plot and the  $I$ - $V$  curve. In Figure 3(a) for example, the maximum resistance (the low frequency x-intercept) of the device at 100% illumination is  $5620 \Omega$ , and this is equivalent to the differential resistance (inverse of the slope) in the  $I$ - $V$  curve at a sample bias of 0 V.

Due to the presence of two partial semicircular arcs, a more complex equivalent circuit model is needed to fit the P3HT:PCBM impedance data. Figure 3(e) presents a model for the P3HT:PCBM device that takes into account the two partial

arcs that appear in the spectra for this device. The high frequency arc is represented by an  $R_1||C_1$  combination and is attributed to the bulk resistance as well as a capacitance that includes both a geometric component and a second component related to photoexcitation events in the film. The larger, low frequency arc is modeled as  $R_2||CPE$  and corresponds to a recombination resistance and chemical capacitance that are associated with the internal charge transfer events at the donor/acceptor interfaces. The use of a constant phase element (CPE) in parallel with a resistor ( $R_2$ ) is often used to describe materials or interfaces that exhibit a distribution of relaxation times,<sup>[37]</sup> including organic light emitting diodes<sup>[43,44]</sup> and photoelectrochemical cells.<sup>[45]</sup> The CPE enables a simpler equivalent circuit model than the previously reported model for OPV devices,<sup>[31]</sup> which included a series of distributed circuit elements with individual time constants. A mathematical description of the distribution function associated with the constant phase element is described in detail in the impedance text by Macdonald,<sup>[37]</sup> but essentially it can be thought of as a non-ideal capacitor possessing a distribution of time constants when put in parallel with a resistor. This  $R_2||CPE$  combination has an impedance  $Z_{CPE}$  of

$$Z_{CPE} = \frac{R_2}{1 + R_2 Q(j\omega)^n} \quad (1)$$

where  $Q$  is the magnitude of the CPE and  $n$  represents an “ideality” factor characteristic of the distribution of relaxation times. A value of  $n = 1$  corresponds to an ideal capacitor, with the  $n$  values in this study ranging from 0.85 to 0.98. The CPE then has an equivalent capacitance  $C_{eq}$  of

$$C_{eq} = \frac{\tau_{avg}}{R_2} = \frac{(R_2 Q)^{1/n}}{R_2} \quad (2)$$

where  $\tau_{avg}$  is the average of the distribution of relaxation times. The variable time constants arise from the inhomogeneous nature of the BHJ system, with a range of charge carrier lifetimes caused by the spatially varying morphology. The determination of an equivalent capacitance makes the CPE more easily relatable to physical processes occurring within the device.<sup>[46]</sup>

As shown in Figure 3, this model provides a high quality fit of the data over a wide range of conditions, with  $C_1$  and CPE having average errors of 1.4% and 5.4%, respectively (as estimated by the fitting software). The impact of illumination intensity and applied bias on the value of the equivalent circuit elements is depicted in Figure 4. Unlike the P3HT-only device, in the BHJ the bulk circuit elements  $R_1$  and  $C_1$  exhibit a dependence on illumination intensity and an increased dependence on bias potential. These effects are likely related to the more highly polarizable nature of the blend film due to internal charge transfer events that occur during photoexcitation. While the details of this response under illumination are not yet fully understood, the value of  $C_1$  in the dark (in the absence of photoexcitation) is simply the geometric capacitance with a value of 2.2 nF. For the given film thickness of 230 nm, this translates into a dielectric constant of 3.5 for the P3HT:PCBM blend, which falls in the range of previously reported values determined by other techniques.<sup>[47,48]</sup> The CPE, on the other hand, is represented by  $C_{eq}$  in Figure 4(a) and increases by 15-fold from dark conditions to 1 sun illumination.  $C_{eq}$  is related to a



chemical capacitance that develops in the blend film as radiant energy is converted to chemical energy, which occurs as charge carriers are generated and stored within the active layer. Chemical capacitance can be derived from different means<sup>[27]</sup> and can be described generally as

$$C_{\mu} = e^2 \frac{\partial n}{\partial E_{Fn}} = e^2 \frac{n_c}{k_B T} \quad (3)$$

where  $C_{\mu}$  is the chemical capacitance per unit volume,  $e$  is the elementary charge,  $n$  is the electron concentration, and  $E_{Fn}$  is the “quasi-Fermi level” generated by excess electrons in the acceptor LUMO. For Boltzmann statistics, which have been observed for these devices at the bias range studied here,<sup>[26,31]</sup> the chemical capacitance can be equivalently defined as a function of the carrier concentration in the conduction band  $n_c$  in Equation 4.

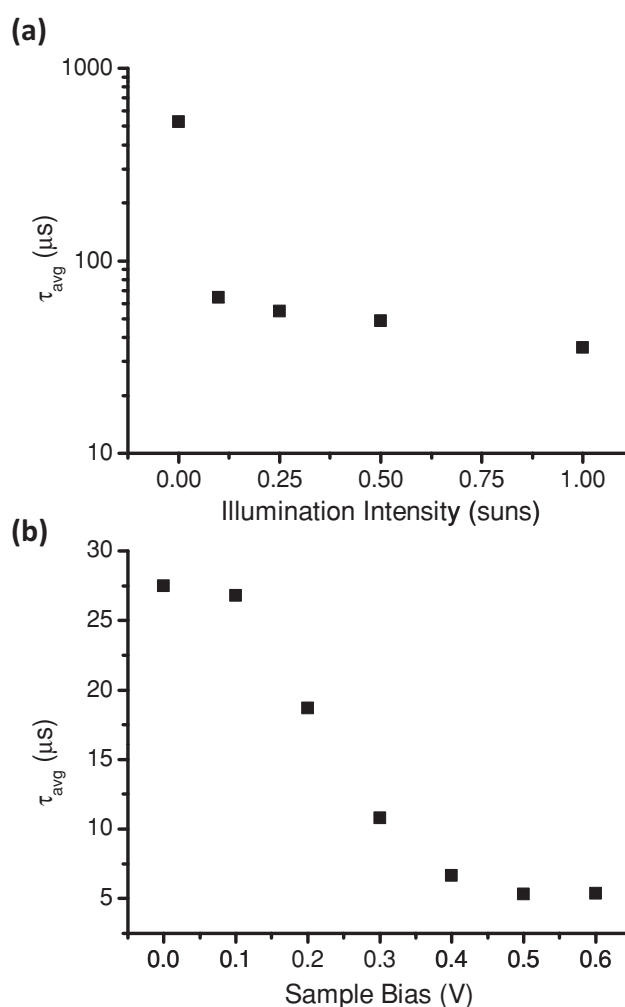
$$n_c = N_c e^{(\mu_n - E_c)/k_B T} \quad (4)$$

In this case,  $N_c$  is the effective density of conduction band states,  $\mu_n$  is the electron chemical potential, and  $E_c$  is the conduction band energy.<sup>[27]</sup>

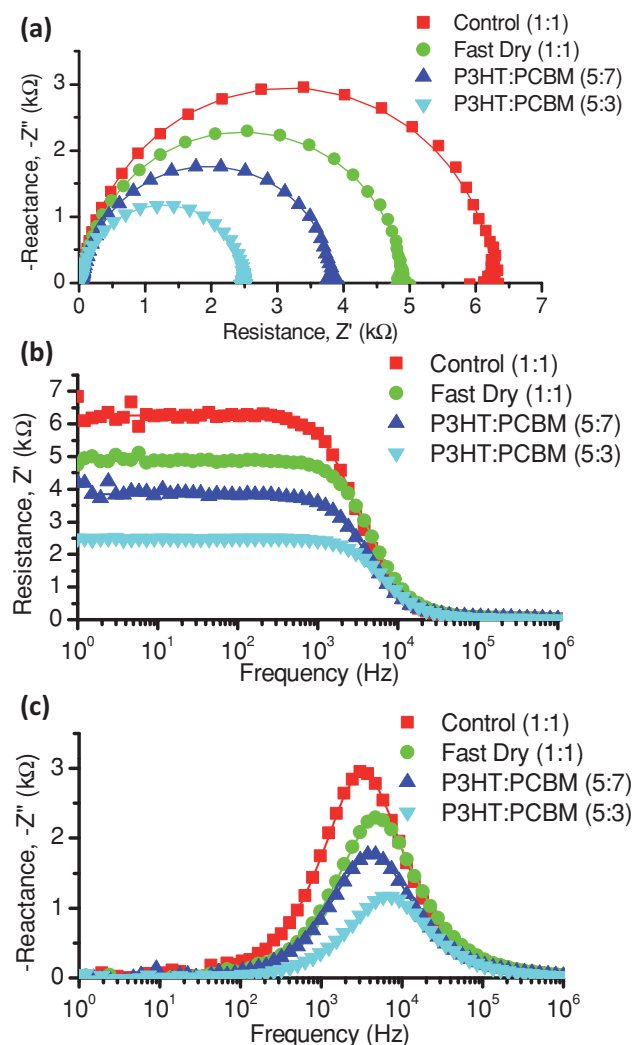
Figure 4(b) shows the response of the equivalent circuit elements to varied sample biases. With increasing bias,  $R_1$  decreases and  $C_1$  increases as charge carriers are injected into the film. Following Boltzmann statistics, the chemical capacitance is expected to scale exponentially with applied bias as an increasing number of charge carriers are stored in the active layer. From 0.0 to 0.6 V, the recombination resistance ( $R_2$ ) exhibits an exponential decay and  $C_{eq}$  an exponential increase, consistent with the predicted inversely proportional and directly proportional responses, respectively, of these factors to the charge carrier density.<sup>[27,32]</sup> Note that the charge carrier density does not vary exponentially with illumination intensity and that  $R_2$  and  $C_{eq}$  therefore do not vary exponentially in that case. As shown in Figure S1, the devices observed here exhibit a small deviation from Boltzmann statistics, which has also been observed in DSSCs and attributed to electron trapping.<sup>[27]</sup> The recombination resistance is a measure of whether the generated charge carriers will be extracted to the electrodes or be lost to bimolecular recombination.<sup>[32]</sup> When  $R_2$  is smaller than the resistance associated with transport through the bulk ( $R_1$ ), charge carriers generated in the active layer are more likely to recombine than to be collected. For sample biases larger than 0.6 V, note that  $R_2$  becomes vanishingly small in comparison to  $R_1$ , and the total measured resistance reduces to the losses in the ITO and PEDOT:PSS ( $R_s$ ) and those in the bulk ( $R_1$ ). At this point, bimolecular recombination losses are expected to be quite large, consistent with the low recombination resistance we observe here. A similar observation can be made from the behavior of the  $I$ - $V$  curve, whose slope (the inverse of the resistance) approaches a constant value at biases greater than  $V_{oc}$  (~0.5 V for these devices).

Although the response of the various circuit elements to bias and illumination would be expected to influence device performance, additional analysis is needed to elucidate their impact on processes within the device. In the case of  $R_2||CPE$ , increasing charge carrier generation consistently correlates with increasing recombination. This effect can be quantified by

calculating the time constants associated with each  $R$ - $C$  combination. The time constant for  $R_1||C_1$  is thought to be related to the electron transit time through the bulk.<sup>[31]</sup> This transit time is essentially insensitive to applied bias and is ~0.74  $\mu$ s at 1 sun illumination for these devices. This charge extraction process is faster than the process associated with  $R_2||CPE$  and is therefore unlikely to be limiting device performance under the conditions studied here. The determination of the average time constant for  $R_2||CPE$ ,  $\tau_{avg}$ , is given in Equation 2, and its response to both illumination and sample bias variation is shown in Figure 5. Note that  $\tau_{avg}$  decreases sharply from dark conditions (where essentially no bimolecular recombination is expected) to 0.1 sun and then decreases exponentially to about 25  $\mu$ s at 1 sun illumination. Increasing the sample bias at constant illumination intensity results in  $\tau_{avg}$  decreasing further until leveling off to ~4  $\mu$ s near 0.5 V.  $\tau_{avg}$  is therefore attributed to the average charge carrier lifetime, which is closely related to bimolecular recombination events in the active layer. This interpretation is consistent with previous reports that have observed an inverse relationship between charge carrier density and bimolecular



**Figure 5.**  $\tau_{avg}$  as a function of (a) illumination intensity at 0.0 V and (b) sample bias at 1 sun illumination.



**Figure 6.** Impedance response of 4 different P3HT:PCBM devices at 0.0 V and 1 sun illumination.

recombination rate at this timescale and illumination intensity.<sup>[23,32,33]</sup> At very high charge carrier densities, the recombination rate and charge carrier density reach a plateau as recombination moves from a trap-limited to a trap-free recombination process,<sup>[34]</sup> which is consistent with our observation in Figure 5(b).

### 2.3. P3HT:PCBM Devices with Varied Composition/Morphology

The present model was then applied to data from devices of varying composition and morphology. The impedance response of P3HT:PCBM (5:7), P3HT:PCBM (5:3), and fast-dry devices are compared to the P3HT:PCBM (1:1) control device in Figure 6. The fast-dry devices had a composition of P3HT:PCBM (1:1) but were dried on a warm hotplate rather than solvent-annealed. This treatment reduced the drying time from about 20 min to 100 sec. Numerous reports have shown that thermal annealing and solvent annealing improve performance by increasing the P3HT crystallinity and increasing the size and the connectivity

**Table 1.** Correlation between device architecture and power conversion efficiency (PCE), short circuit current density ( $J_{sc}$ ),  $\tau_{avg}$ , and peak reactance frequency ( $f$ ).

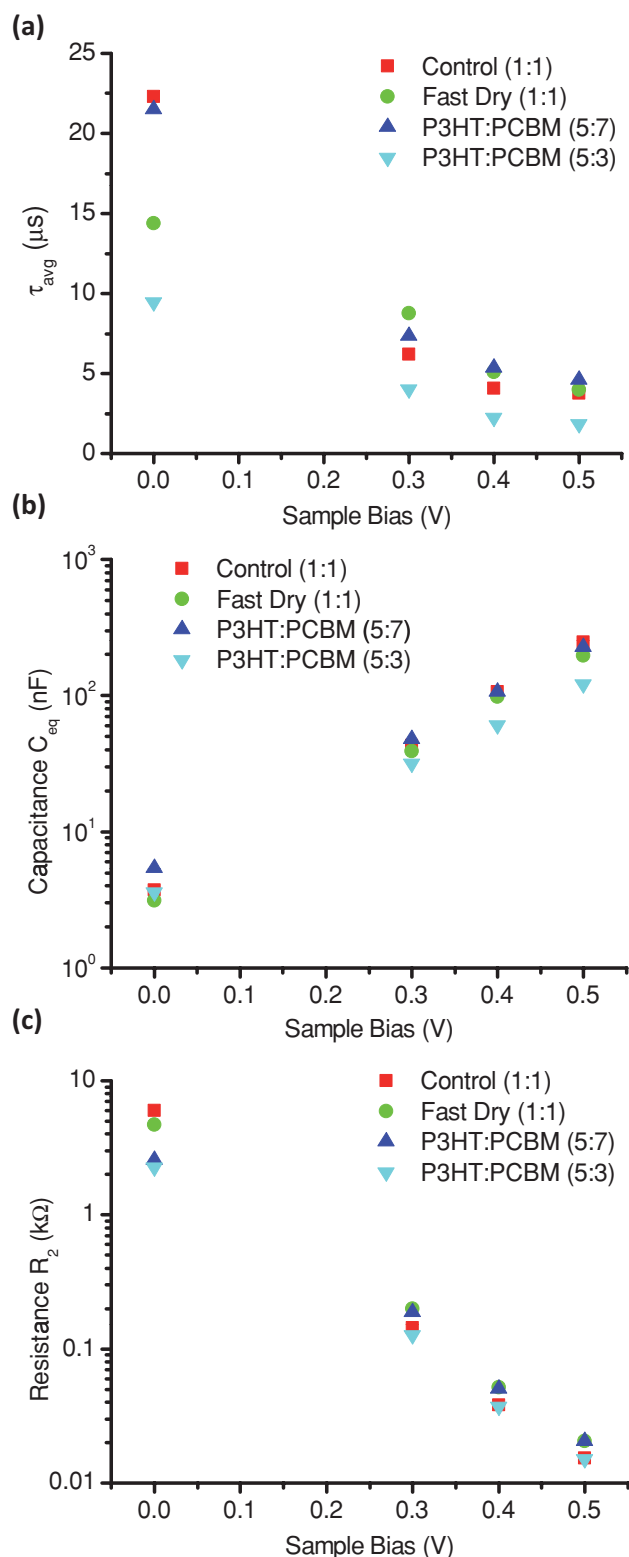
|                 | PCE (%) | $J_{sc}$ (mA/cm <sup>2</sup> ) | $\tau_{avg}$ (s) | $f$ (Hz) |
|-----------------|---------|--------------------------------|------------------|----------|
| Control         | 2.5     | 8.6                            | 22               | 3020     |
| P3HT:PCBM (5:7) | 2.4     | 8.4                            | 21               | 3780     |
| Fast Dry        | 2.2     | 7.4                            | 14               | 4720     |
| P3HT:PCBM (5:3) | 2.0     | 8.0                            | 7.8              | 5900     |

of the P3HT domains within the BHJ network.<sup>[11,12,18]</sup> Fast-dry devices would therefore be expected to have less interconnected domains in comparison to the control devices. The impact of varying the P3HT:PCBM ratio is less clear, and the relationship between active layer morphology and composition is complex. Although an optimal ratio of 1:1 is commonly observed,<sup>[7,49,50]</sup> other groups have reported different optimal compositions near 1:1.<sup>[8,12,51]</sup> Electron Microscopy studies have shown that non-optimal P3HT:PCBM compositions impact morphology by reducing P3HT crystallinity within the active layer and causing unfavorable vertical phase segregation.<sup>[12]</sup>

Representative impedance data, both in the form of Cole-Cole plots as well as the frequency dependence of the impedance (Bode plots) for each device architecture at short circuit conditions, are shown in Figure 6. Note that  $\tau_{avg}$  can be correlated with the frequency at which the reactance reaches its peak value,<sup>[25]</sup> and as shown in Figure 6(b), there is a clear correlation between  $\tau_{avg}$  (i.e., frequency) and device structure. This is made evident in Table 1, which further reveals a correlation exists between  $\tau_{avg}$  and device performance. Larger  $\tau_{avg}$  values (from 7.8  $\mu$ s to 22  $\mu$ s) lead to higher power conversion efficiencies (from 2.0% to 2.5%).

The above correlation between  $\tau_{avg}$  and device structure suggests that bimolecular recombination plays an important role in these devices under short circuit conditions, a point which has recently been in some dispute. For example, recent work using transient absorption spectroscopy reported a weak correlation between bimolecular recombination and composition under short circuit conditions and suggested that short circuit current density ( $J_{sc}$ ) should be a stronger predictive tool since geminate recombination dominates under these conditions.<sup>[49]</sup> A separate study reports, however, that up to 17% of charge carriers can be lost to bimolecular recombination under short circuit conditions.<sup>[35]</sup> Note that we observe a correlation between lifetime at 0.0 V and efficiency in both the case of varied composition and varied morphology at a uniform composition, which supports the argument that bimolecular recombination plays an important role under these conditions.

We also investigated the relationship between the equivalent circuit parameters at biases larger than 0.0 V as shown in Figure 7. Moving away from short circuit conditions, the control, fast dry, and P3HT:PCBM (5:7) architectures all approach a lifetime near 4  $\mu$ s, while the lifetime of the P3HT:PCBM (5:3) device remains significantly shorter. In this case, the active layer film was about 20% thinner than in the other devices. However, the significantly lower values of both  $C_{eq}$  and  $R_2$  for the 5:3 blend suggest that this is a non-optimal morphology.



**Figure 7.** (a)  $\tau_{avg}$ , (b) chemical capacitance, and (c) recombination resistance as a function of sample bias at 1 sun illumination.

This conclusion is further supported by AFM phase images provided in the Figure S2, which appear to show the greatest extent of phase segregation in this blend. As shown in Figure 7,

the recombination resistance of the 5:3 blend is  $\sim 1/3$  of the value for the control device at 0.0 V. Although this difference narrows with increasing bias, the 5:3 blend develops a capacitance deficit relative to the control that reaches 50% at 0.5 V. While increasing the thickness of the 5:3 blend film would likely improve the performance, we can conclude that its performance would not equal that of the 1:1 blend. This analysis of the chemical capacitance and recombination resistance highlights their utility in device optimization by providing additional information beyond lifetime alone.

### 3. Conclusions

In this work, we have characterized the impedance response of P3HT:PCBM organic photovoltaic devices over a range of applied biases and illumination conditions. The devices can be modeled by an equivalent circuit consisting of resistors, a capacitor, and a constant phase element. The model takes into account both the geometrical capacitance as well as the chemical capacitance and recombination resistance related to charge carrier accumulation and bimolecular recombination in the active layer. Furthermore, average charge carrier lifetimes can be extracted from the equivalent circuit model and correlated with device performance. In our analysis at the short circuit condition, the average electron lifetime was found to vary from 7.8 to 22  $\mu s$  for devices with power conversion efficiencies ranging from 2.0 to 2.5%. In addition, the chemical capacitance and recombination resistance can be analyzed individually, and can inform process changes geared toward increasing power conversion efficiency. Overall, this work establishes impedance spectroscopy as an effective technique for characterizing operational BHJ OPVs and thus holds promise for informing efforts to optimize the morphology of new bulk heterojunction materials systems.

## 4. Experimental Section

### 4.1. Device Preparation

Indium tin oxide (ITO) coated glass slides ( $1.5'' \times 1.0''$ , sheet resistance =  $\sim 10 \Omega/\square$ , RMS roughness = 3–5 nm) were purchased from Colorado Concept Coatings. To create an area where the completed devices could be later contacted, a  $1/4''$  strip of ITO was etched from opposite edges of the slides by affixing electrical tape to the center of the slides and then immersing them in concentrated HCl for 20 min. After thoroughly rinsing the slides in de-ionized water, they were then cleaned by sonicating sequentially for 15 min in each of the following solvents: dilute Alconox<sup>TM</sup> aqueous detergent, de-ionized water, hexanes, 2-propanol, acetone, and methanol. The slides were blown dry with nitrogen between each solvent and were stored in a closed container under  $N_2$  until use.

Immediately before device fabrication, the cleaned ITO slides were exposed to ozone for 20 min in a Novascan PSD Pro Series Digital UV Ozone System. To deposit the interfacial layer poly(3,4-ethylenedioxythiophene): poly(styrene

sulfonate) (PEDOT:PSS), Clevios™ PH (H.C. Starck) was dropped onto each slide through a 13-mm dia. PVDF syringe filter with 0.45  $\mu\text{m}$  pores (Millex®-GV) and then spin-coated at 3500 rpm for 45 sec. The Clevios™ PH was stored at 4 °C and then allowed to warm to room temperature before use. The resulting films were annealed at 140 °C on a hot plate for 15 min. to drive off the water, and were determined by profilometry (KLA Tencor P15 Stylus Profiler) to be ~45 nm thick. To prevent water adsorption into the PEDOT:PSS films, the substrates were immediately transferred to a N<sub>2</sub> glove box (<1 ppm O<sub>2</sub>, <1 ppm H<sub>2</sub>O) after annealing.

For the active layers, 3 different P3HT:PCBM solutions were prepared in the nitrogen glove box. For the control devices, P3HT (electronic grade, 98% regioregular, Rieke metals) and PCBM (>99.95%, American Dye Source) were dissolved in anhydrous 1,2-dichlorobenzene (Sigma-Aldrich) to yield P3HT:PCBM (1:1) solutions with each component having a concentration of 18 mg/mL. All materials were used as received. Additional solutions were prepared with P3HT:PCBM (5:7), P3HT:PCBM (5:3), and P3HT-only (no PCBM) compositions, each having a P3HT concentration of 18 mg/mL. All of the solutions were stirred overnight in an oil bath at 60 °C. The next day, each solution was dropped onto a PEDOT:PSS-coated ITO substrate through a 13-mm dia. PTFE syringe filter with a 0.22- $\mu\text{m}$  pore size (Millex®-FH) and then spin-coated at 600 rpm for 1 min. The control device, the P3HT-only device, and those with P3HT:PCBM (5:7) and (5:3) compositions were solvent-annealed by allowing them to dry for ~20 min. in separate Petri dishes. A fifth device architecture, which will be called “fast-dry,” was prepared by spin-coating the control mixture under the same conditions but drying it on a warm hot plate (~35 °C) so that that film dried in about 100 sec. After cleaning the organic layers from the glass edges of each substrate with a chlorobenzene-soaked cotton swab, each film was annealed in an oven at 120 °C for 10 min. Finally, top metal electrodes were deposited by thermally evaporating ~100 nm Al through a shadow mask onto each film. Following metal deposition, each substrate had 10 devices with individual active areas of 0.165 cm<sup>2</sup>.

## 4.2. Device Characterization

Without removing the devices from the glove box, current-voltage (*I*-*V*) curves (not shown) were recorded using a 300 W Oriol 81160 solar light simulator equipped with a Xe arc lamp and an AM1.5G filter. Devices were illuminated with 100 mW/cm<sup>2</sup> radiation while the bias was swept from -1.0 V to 1.0 V. For the impedance measurements, each device was placed in a sample holder that was purged continuously with nitrogen after removal from the glove box. Impedance measurements were conducted with a Novocontrol Broadband Dielectric Spectrometer equipped with a 4-Wire Impedance Test Interface (in a 2-probe configuration). The impedance response was measured over the range of 1 Hz to 1 MHz with an oscillation amplitude of 10 mV. The impedance was recorded under dark and illuminated conditions over a DC bias range of 0.0 V to 0.75 V. Devices were illuminated with a halogen lamp at ~100 mW/cm<sup>2</sup>. For each device, the 100% illumination

intensity (nominally 100 mW/cm<sup>2</sup>) was set by adjusting the lamp such that the device exhibited the same short circuit current (*I*<sub>sc</sub>) that was measured under AM1.5G illumination. Measurements at reduced illumination were made by using neutral density filters. Equivalent circuit modeling was optimized with EIS Spectrum Analyser software.<sup>[52]</sup>

## Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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# ADVANCED ENERGY MATERIALS

## Supporting Information

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In Situ Characterization of Lifetime and Morphology in  
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by Impedance Spectroscopy

*Benjamin J. Leever , Christopher A. Bailey , Tobin J. Marks , \* Mark  
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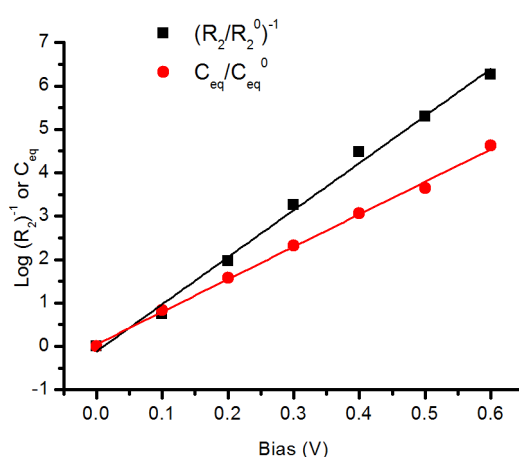


Figure S1.  $C_{eq}$  and  $R_2$  as a function of bias.  $C_{eq}^0$  denotes the value of  $C_{eq}$  at 0.0 V and  $R_2^0$  denotes the values of  $R_2$  at 0.0 V. As described in Ref. 27, for Boltzmann statistics, the slope is expected to be  $(e/k_B T)^{-1} = 0.026$ . We find a slope for  $R_2$  of 0.092 and for  $C_{eq}$  of 0.13. The value determined for DSSCs (in Ref. 27) was reported to be 0.10.

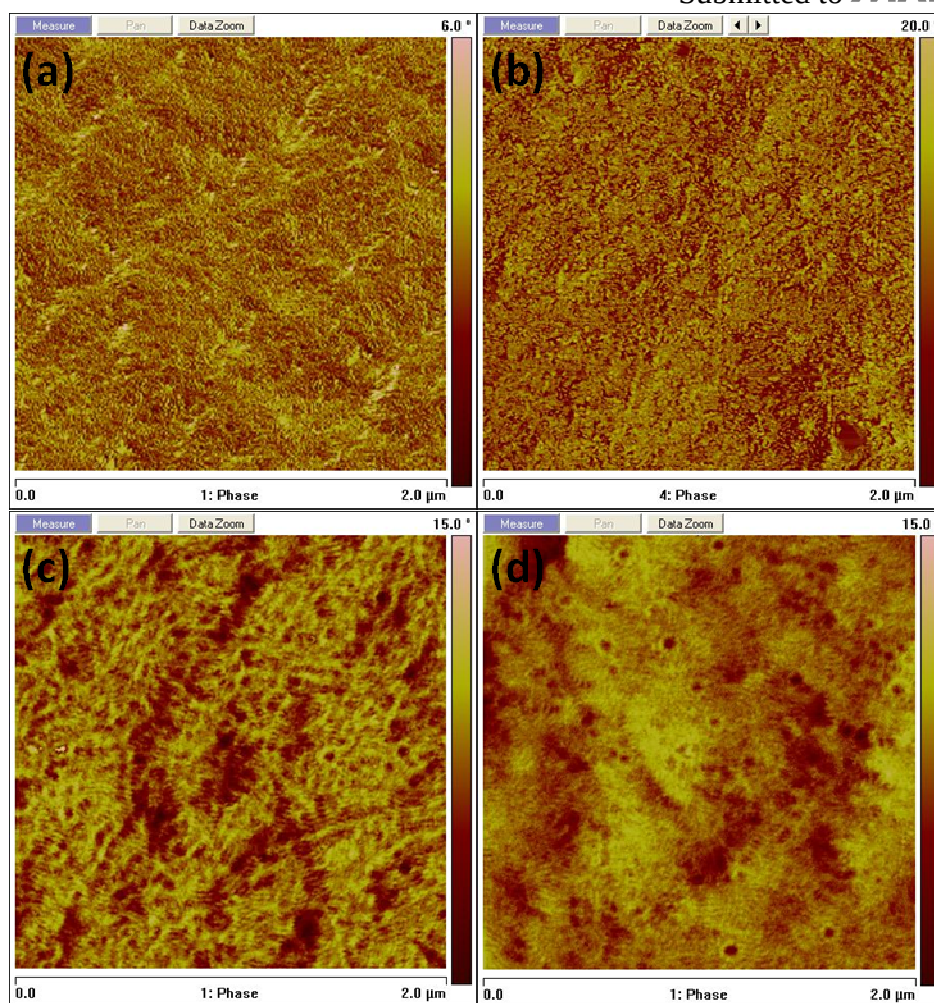


Figure S2. AFM Phase images of P3HT:PCBM blend films. Blends are (a) 1:1, (b) 5:7, (c) 5:3, and (d) fast dry. Note that apparent phase segregation (as indicated by primarily yellow or red areas) increases significantly for the 5:3 and fast dry blends as compared to the 1:1 and 5:7 blends.

### Experimental Procedure for AFM

Atomic force microscopy was conducted using a Veeco Nanoscope V instrument. The microscope was operated in tapping mode using Nanosensors PPP-NCHR AFM probes with radii of curvature of  $\sim 7$  nm.